

5. SOLUTIONS OF 1-D STEADY CONDUCTION EQUATION

(1) Introduction

$$\text{Steady} \Rightarrow \frac{\partial}{\partial t} = 0, \text{ 1-d} \Rightarrow \frac{\partial}{\partial y} = 0, \frac{\partial}{\partial z} = 0$$

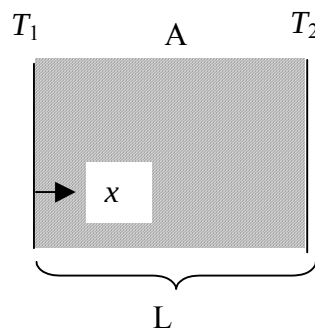
$$\text{The governing equation thus reduces to } \frac{d}{dx} \left(k \frac{dT}{dx} \right) = 0.$$

$$\text{If thermal conductivity is constant, } \frac{d^2 T}{dx^2} = 0 \text{ (linear profile)}$$

Note: For prescribed temperatures at the two boundaries, T is independent of material properties. What is going on?

(2) Thermal resistance

(CS1)



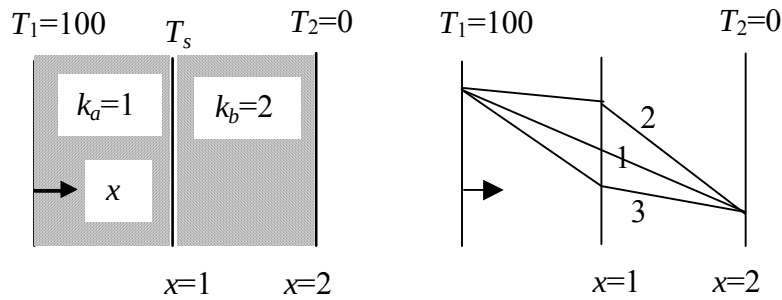
$$T(x) = C_1 x + C_2 : \text{two arbitrary constants because of integration}$$

$$\text{Applying B.C., } T(0) = T_1 \text{ and } T(L) = T_2, \quad T(x) = \frac{T_2 - T_1}{L} x + T_1.$$

$$\text{By Fourier's law, } q = -kA \frac{dT}{dx} = -kA \frac{T_2 - T_1}{L} = \frac{T_1 - T_2}{L/kA}$$

- Temperature profile is linear
- HT rate is linearly proportional to ΔT and inversely proportional to L/kA .

(CS2)



Physically, temperature profile should be 3. Why? Because temperature is more uniform in the region with high thermal conductivity. Fourier's law and energy balance also indicate that the temperature gradient in the region of high thermal conductivity has to be small in order to maintain a constant heat flux.

$$q = -kA \frac{dT}{dx} = \text{const.} \text{ Therefore, } k \downarrow \text{ then } \frac{dT}{dx} \uparrow.$$

In $\frac{d}{dx} \left(k \frac{dT}{dx} \right) = 0$, k now depends on x . The two layers need to be considered separately as the thermal conductivity is constant for each layer.

$$\text{BC's } T(0) = T_1, \quad T(2) = T_2, \quad q = -k_a A \frac{dT}{dx} \Big|_{x=1^-} = -k_b A \frac{dT}{dx} \Big|_{x=1^+} \Rightarrow T_s = 33.3^\circ\text{C}$$

(3) Electrical circuit analogy

Is there an easier way for the above analysis? Electrical circuit analogy or thermal network method.

Analogy between heat transfer and electricity

$$q = kA \frac{T_1 - T_2}{L} = \frac{T_1 - T_2}{(L/kA)} \rightarrow q = \frac{\Delta T}{R_t} \Leftrightarrow i = \frac{\Delta V}{R}$$

Definition: thermal resistance

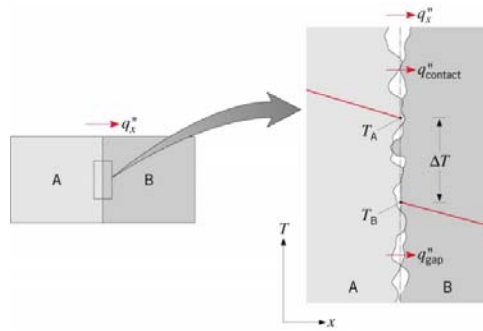
$$R_t \equiv \frac{\Delta T}{q}$$

Definition: Overall heat transfer coefficient

$$q = UA\Delta T = \frac{\Delta T}{R_{tot}} \quad UA \equiv \frac{1}{R_{tot}}$$

Summary (Table 3.3)

- Thermal resistance of a plate: $\frac{L}{kA}$ [K/W]
- Thermal resistance of a cylinder: $\frac{\ln(r_2/r_1)}{2\pi Lk}$
 $\leftarrow \frac{1}{r} \frac{d}{dr} \left(kr \frac{dT}{dr} \right) = 0 \quad q_r = -kA \frac{dT}{dr} = -k(2\pi rL) \frac{dT}{dr} = \frac{2\pi Lk(T_1 - T_2)}{\ln(r_2/r_1)}$
- Thermal resistance of a sphere: $\frac{(1/r_1) - (1/r_2)}{4\pi k}$
 $\leftarrow \frac{1}{r^2} \frac{d}{dr} \left(kr^2 \frac{dT}{dr} \right) = 0 \quad q_r = -kA \frac{dT}{dr} = -k(4\pi r^2) \frac{dT}{dr} = \frac{4\pi k(T_1 - T_2)}{(1/r_1) - (1/r_2)}$
- Convection thermal resistance: $\frac{1}{hA}$ [m²·K/W]
- Thermal contact resistance: $R''_{t,c} \equiv \frac{\delta T}{q''_x} = \frac{T_A - T_B}{q''_x}$ (3.1.4)



➤ Equivalent thermal circuit (Fig. 3.3)

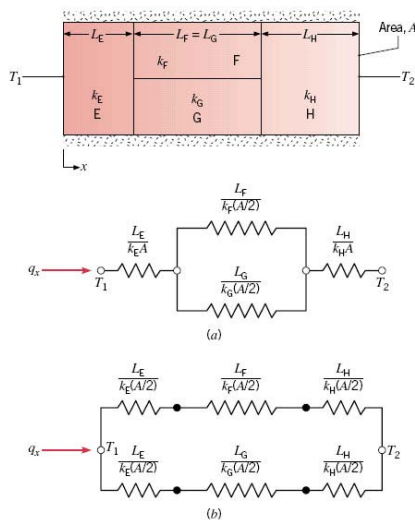


FIGURE 3.3 Equivalent thermal circuits for a series-parallel composite wall.

➤ Critical thickness of insulation layer (Example 3.6)

(4) 1-d steady conduction (variable k and A , internal heat generation)

Variable thermal conductivity and area

$$q = -k(T)A(x)\frac{dT}{dx}$$

In 1-d steady state HT, heat transfer rate q is conserved:

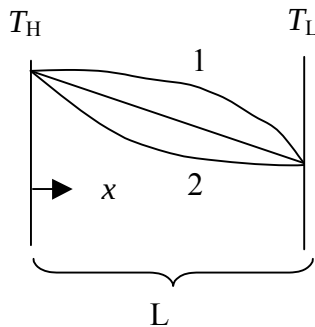
High thermal conductivity $\Rightarrow$ small temperature gradient

Large cross sectional area $\Rightarrow$ small temperature gradient

Analysis

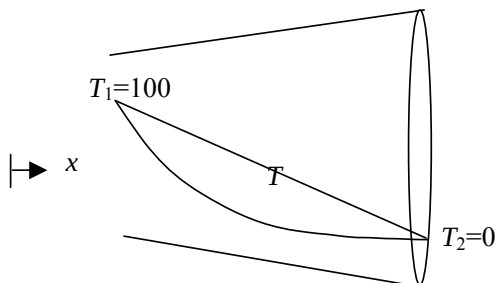
Direct integration of Fourier's law: $\dot{Q} \int \frac{1}{A(x)} dx = - \int k(T) dT$

(Ex1) Thermal conductivity $k = aT + b$ ($a, b > 0$)



Profile 1 is the correct answer.

(Ex2) $A(x) = x^2$



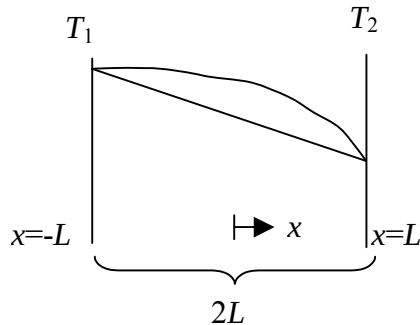
$$q = -kx^2 \frac{dT}{dx} \Rightarrow -\frac{q}{k} x^{-2} dx = dT \Rightarrow \int C_1 x^{-2} dx = \int dT \Rightarrow T = C_1 x^{-1} + C_2$$

Apply BC's to determine unknown constants.

(5) Conduction with thermal energy generation

Thermal network method inapplicable $\Rightarrow$ Solve the governing equation directly

Plane wall (1-d, steady-state, constant properties)



$$\frac{d^2 T}{dx^2} + \frac{\dot{q}}{k} = 0$$

BC's: $T(-L) = T_1$ and $T(L) = T_2$

$$T(x) = \frac{\dot{q}L^2}{2k} \left(1 - \frac{x^2}{L^2} \right) + \frac{T_2 - T_1}{2} \left(\frac{x}{L} \right) + \frac{T_1 + T_2}{2} \quad (3.41)$$

– $T(x)$ is not linear but parabolic

– Maximum temperature? $\left. \frac{dT}{dx} \right|_{x=x_m} = 0$ at $x_m = \frac{k(T_2 - T_1)}{2L\dot{q}}$ (As $\dot{q} \rightarrow \infty$,

$$x_m \rightarrow 0)$$

– If $T_1 = T_2 = T_s$, $T(x) = -\frac{\dot{q}L^2}{2k} \left(1 - \frac{x^2}{L^2} \right) + T_s$ (maximum temperature at $x=0$)

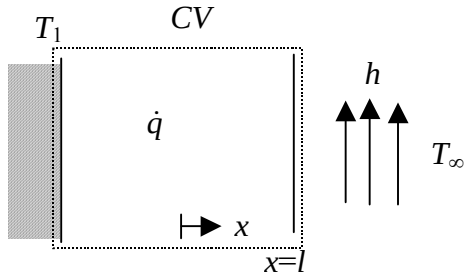
– Heat flux at any arbitrary location is calculated by $q'' = -k \frac{dT}{dx}$

$$q''(x) = -k \frac{dT}{dx} = -k \left(\frac{\dot{q}x}{k} + \frac{T_2 - T_1}{2L} \right) = -\dot{q}x - \frac{k(T_2 - T_1)}{2L}$$

– Note that the heat transfer rate is not a constant any more. It varies as a function of x .

- Overall energy balance? $\dot{q}V = ?$

(Ex)



Q. $T(x)$? $T_s = T(l/2)$?

Solve the ODE and apply the BC's

Or use Eq. (3.41) with BC's

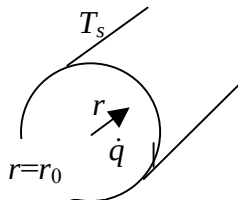
$$q''\left(-\frac{l}{2}\right) = -k \frac{dT}{dx}\bigg|_{x=-\frac{l}{2}} = 0 \quad \text{and} \quad q''\left(\frac{l}{2}\right) = -k \frac{dT}{dx}\bigg|_{x=\frac{l}{2}} = h \left[T\left(\frac{l}{2}\right) - T_\infty \right]$$

Alternative approach to get T_s :

$$\text{Energy balance } \dot{q}V = \dot{q}lA = hA(T_s - T_\infty) \Rightarrow T_s$$

[Cylinder of radius r_0]

When a cylinder has uniform thermal energy generation, the temperature distribution is obtained by solving the governing equation.



$$\text{Governing equation } \frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) + \frac{\dot{q}}{k} = 0$$

BC's $\left. \frac{dT}{dr} \right|_{r=r_0} = 0$ and $T(r_0) = T_s$

$$T(r) = \frac{\dot{q}r_0^2}{4k} \left(1 - \frac{r^2}{r_0^2} \right) + T_s$$

The radial heat flux is not a constant but a function of r : $q_r''(r) = -k \frac{dT}{dr} = -k \left(\frac{\dot{q}r}{2k} \right) = \frac{\dot{q}r}{2}$.

Heat flux increases linearly with r .

Temperature profile?

[Sphere of radius r_0]

Similarly, $T(r) = \frac{\dot{q}r_0^2}{6k} \left(1 - \frac{r^2}{r_0^2} \right) + T_s$ ($\leftarrow$ Try to obtain this.)